

Experiment No - 3

Aim: Write An ALP for memory block transfer.

Objective: To understand to use, and to develop a simple assembly language program to transfer a block of data from one memory location to another using 8086 instructions.

Hardware:

Device Name: Computer Think-Centre.

Hardware Model: Lenovo Think-Centre neo 50t Gen 3

Processor: 12th Gen Intel® Core™ i5 - 12400 X 12

Disk Capacity: 256.1 GB

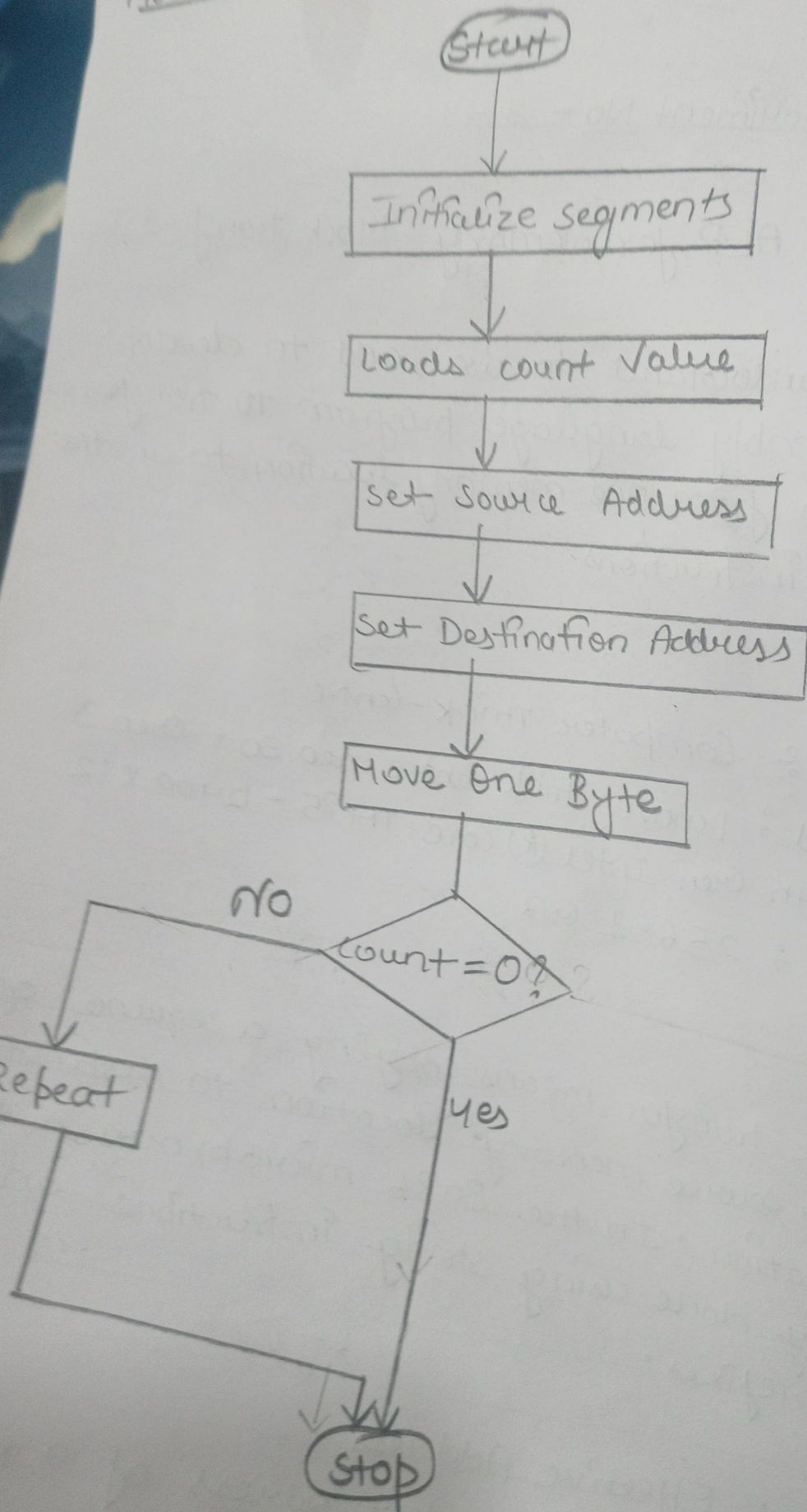
Theory:

Memory block transfer means copying a sequence of bytes from a source memory location to destination memory location. In the 8086 microprocessor, this is efficiently done using string instructions and repetition prefixes.

LEA — Load Effective Address

LEA is used to load the offset address of a memory variable into a register.

Flowchart



Important Points

- Holds address, not the actual data.
- Useful for accessing memory locations.
- Helps initialize index registers.
- Used in array and string operations.
- Improves programmers' effectiveness.

REP - Repeat Prefix

- REP is used to repeat a string operation multiple times automatically.

Important Points

- Works with string instructions.
- Uses count value stored in a register.
- Repeats operation until count becomes zero.
- Eliminates need for manual looping.
- Speeds up block data transfer.

MOVSB - Move String Byte

MOVSB transfers one byte from source memory to destination memory automatically.

Important Points

- Copies byte-by-byte data.
- Source and destination addresses update automatically.
- Used for block transfer operations.
- Works efficiently with REP.
- Supports sequential memory copying.

SYNTAX

- MODEL SMALL
- STACK
- DATA

SRC DB 01H, 02H, 03H

DST DB 3 DUP (?)

- CODE

MOV AX, @DATA

MOV DS, AX

MOV ES, AX

; Transfer operation

MOV AH, 4CH

INT 21H

END.

Example (Block Transfer)

LEA SI, SRC

LEA DI, DST

MOV CX, 3

REP MOVSB

EQU - Equate Directive

EQU assign a constant value to a symbolic name.

Important Points

- Used to define constants
- Improves readability of program
- Prevents use of hard-coded value
- Value cannot be changed during execution.
- Helps in structured programming

ORG - Origin Directive

ORG specifies the starting address of the program in memory.

- Set program loading address
- Important for correct memory mapping
- Used in system-level programming
- Helps control program placement
- Ensure proper execution.

Algorithm

Step 1) Start

Step 2) Initialize data segment extra segment

Step 3) Load count value in count register.

Step 4) Initialize destination memory location.

Step 5) Load Source memory address.

Step 6) Move bytes from source to destination repeatedly until count become zero.

Step 7) Stop.

Conclusion.

The experiment demonstrated how a block of data can be transferred from one memory location to another using string instructions of the 8086 microprocessor. Instructions like REP and MOVSB make the transfer efficient and automatic. Memory block transfer is an important operation in data handling and system programming.